Self-Reported Youth and Adult Exposure to Alcohol Marketing in Traditional and Digital Media: Results of a Pilot Survey

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Background: Alcohol marketing is known to be a significant risk factor for underage drinking. However, little is known about youth and adult exposure to alcohol advertising in digital and social media. This study piloted a comparative assessment of youth and adult recall of exposure to online marketing of alcohol.

Methods: From September to October 2013, a pilot survey of past 30-day exposure to alcohol advertising and promotional content in traditional and digital media was administered to a national sample of 1,192 youth (ages 13 to 20) and 1,124 adults (ages ≥21) using a prerecruited Internet panel maintained by GfK Custom Research. The weighted proportions of youth and adults who reported this exposure were compared by media type and by advertising and promotional content.

Results: Youth were more likely than adults to recall exposure to alcohol advertising on television (69.2% vs. 61.9%), radio (24.8% vs. 16.7%), billboards (54.8% vs. 35.4%), and the Internet (29.7% vs. 16.8%), but less likely to recall seeing advertising in magazines (35.7% vs. 36.4%). Youth were also more likely to recall seeing advertisements and pictures on the Internet of celebrities using alcohol (36.1% vs. 20.8%) or wearing clothing promoting alcohol (27.7% vs. 15.9%), and actively respond (i.e., like, share, or post) to alcohol-related content online.

Conclusions: Youth report greater exposure to alcohol advertising and promotional content than adults in most media, including on the Internet. These findings emphasize the need to assure compliance with voluntary industry standards on the placement of alcohol advertising and the importance of developing better tools for monitoring youth exposure to alcohol marketing, particularly on the Internet.

Key Words: Alcohol, Marketing, Youth, Internet, Social Media.

EXCESSIVE ALCOHOL USE is responsible for about 4,300 deaths among persons under age 21 years in the United States each year, and underage drinking cost the U.S. $24.3 billion in 2010 (Centers for Disease Control and Prevention, 2015; Sacks et al., 2015). Underage drinking is also associated with many health risks, including smoking, physical fighting, and high-risk sexual activity (Miller et al., 2007). The prevention of excessive alcohol use, including underage drinking, is therefore an important public health priority.

At least 25 longitudinal studies have found that youth exposure to alcohol marketing in various forms increases the likelihood that young people will begin drinking, or if already drinking, progress to more hazardous drinking patterns (Anderson et al., 2009; Jernigan et al., 2017). Nonetheless, many studies have documented that young people are overexposed (i.e., have greater exposure to alcohol advertising per capita than adults) to alcohol advertising on television, radio, and in magazines (Center on Alcohol Marketing and Youth, 2010, 2011, Jernigan et al., 2013; Ross et al., 2017). However, less is known about youth and adult exposure to alcohol advertising in digital and social media (i.e., on the Internet) despite rapid growth in alcohol advertising in these media (Federal Trade Commission, 2014; Jernigan and Rushman, 2014).

A major challenge in assessing exposure to alcohol advertising on the Internet is that unlike traditional media, which can be assessed using commercially available date sets such as Nielsen (Jernigan and Ross, 2010), there is currently no comprehensive data source available to the public for assessing exposure to alcohol advertising on the Internet. This is due, in part, to the way alcohol advertising is purchased and presented to Internet users. The Internet provides a highly

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customized advertising medium, with detailed browsing histories available to advertisers, enabling them to finely target promotions to different users. As a result, the advertising experience of people viewing content on the Internet is unique to each viewer and may vary considerably depending on the content seen on different subpages of an Internet site, making it difficult to assess advertising exposure on a population basis using traditional methods.

In contrast, most viewers of nationally broadcast network or cable television programs, listeners to over-the-air radio stations, and readers of national editions of magazines see, hear, or read the same commercials embedded within the programming or in print. Commercially available advertising research data for these traditional media are structured to take advantage of the shared advertising experiences of people using the medium. Audience data for a radio station, television program, or magazine are compiled using a combination of survey methods and automated technologies. These audience data can then be matched to advertising occurrence data to estimate the audience that was exposed to an advertisement (Center on Alcohol Marketing and Youth, 2010, 2011; Jernigan and Ross, 2010; Ross et al., 2017).

The online advertising experience is further complicated by the growing use of a significant subset of digital media, namely social media platforms (e.g., Facebook), for advertising. Within social media, people may be exposed to customized advertising in their newsfeed (the term for the customized updates to a user's home page on Facebook based on what people with whom they have linked or "friended" on Facebook are posting, and interspersed with advertising served to the users by Facebook's content algorithms) or may interact with branded products directly on pages set up for this purpose by advertisers (Hastings and Sheron, 2013). Further, readers/viewers may become active agents in the promotion of products by liking, sharing, or posting branded commercial messages, thereby endorsing the message among their peers and, whether aware of it or not, participating in "viral marketing" (Chester et al., 2010; Jernigan and Rushman, 2014).

In light of the methodological challenges involved in trying to assess exposure to alcohol advertising on the Internet and in social media, the aim of this study was to pilot a comparative assessment of youth and adult recall of exposure to alcohol marketing in traditional media (i.e., television, radio, magazines, and billboards or other outdoor signage) and on the Internet, including exposure to and engagement in marketing in social media. In so doing, we specifically sought to answer the following research questions: (i) Is assessment of youth and adult exposure to alcohol advertising on the Internet feasible using an Internet panel?; (ii) What proportion of youth (aged 13 to 20 years) and adults (aged 21 and above) recall being exposed to alcohol marketing on traditional media and on the Internet?; (iii) Of those who were exposed to this advertising, how frequently do they recall being exposed to it?; (iv) What proportion of youth and adults report actively engaging with alcohol marketing in social media, including viewing, liking, sharing, or posting?

MATERIALS AND METHODS

Design Overview

A national sample of 1,192 underage youths, ages 13 to 20, and 1,124 adults, age 21 or older was selected from a prerecruited Internet panel maintained by GfK Custom Research (Palo Alto, CA) (Knowledge Networks, 2012). During September to October 2013, study participants completed an online, self-administered survey that asked about exposure to and engagement with alcohol advertising and promotional content in traditional and digital media. The Institutional Review Board of the Johns Hopkins Bloomberg School of Public Health approved the protocol. A Certificate of Confidentiality was obtained from the National Institutes of Health to help protect the confidentiality of the panelists' responses.

Sample

GfK Custom Research maintains a prerecruited Internet panel of approximately 50,000 adults (including young adults ages 18 to 20) and approximately 3,000 teens (ages 13 to 17) who have consented (and whose parents have provided permission) to be a panel member (KnowledgePanel®). The panel was created using a national probability sample. Households were recruited through both random digit dialing (RDD) and address-based sampling (ABS), which involves probability sampling of addresses from the U.S. Postal Service's Delivery Sequence File (Knowledge Networks, 2012). The sampling frame covers 97% of U.S. households (Knowledge Networks, 2012), but does not include institutionalized populations. To ensure adequate representation of hard-to-reach demographic groups, GfK Custom Research oversamples certain minority groups and offers laptops and Internet connectivity to those without equipment and access. Panel members are invited through email to complete surveys on average 4 times a month (Knowledge Networks, 2012). Panel members with Internet access receive incentive points (redeemable for cash) per survey completed; panel members who were initially without a home computer and Internet access receive these in lieu of incentive points.

Study participants aged 18 years and older were directly contacted to obtain informed consent to participate in the study. If they consented, they were provided with a secure link to the survey. For potential participants age 13 to 17 years, parental consent was sought via email. If obtained, these potential participants were asked to consent to participating in the survey as well. If they consented, they were provided with a link to the survey. Because of the brevity of the survey (<10 minutes), no financial incentive was offered for adults; however, a $5 gift was credited to the 13- to 17-year old panel members after survey completion.

Survey Administration

GfK pilot tested the survey to confirm survey functionality and duration. The survey was then administered to respondents from September 25 to October 14, 2013. As media consumption generally declines during the summer, this time period was selected to better characterize typical media exposure throughout the year.

Weighting Procedures

Knowledge Networks applied statistical weighting adjustments to account for selection biases, yielding a sample that more closely represents the general population (DiSogra, 2009). These weights accounted for the different selection probabilities associated with the RDD- and ABS-based samples, the oversampling of minority communities, nonresponse to panel recruitment, and panel attrition. Poststratification adjustments were based on demographic distributions from the most recently available Current Population Survey conducted by the U.S. Census Bureau (2013). The poststratification weights adjusted for gender, age, race/ethnicity, education, census region, household income, and metropolitan area.

Validity of Methodology

Validation studies have concluded that behavioral data obtained from the KnowledgePanel compare closely with estimates derived from more traditional survey techniques, such as national household, telephone, or in-person surveys (Bethell et al., 2004; Chang and Krosnick, 2009; Heeren et al., 2008; Yeager et al., 2011). Moreover, the Knowledge Networks panel yields results that meet or exceed the accuracy of standard RDD telephone surveys or in-person interviews.

Measures

Participants were asked to report if they had seen or heard any alcohol advertising in any of 5 different media venues during the previous 30 days. Specifically, participants answered "Yes"/"No" to the question "In the last 30 days, have you seen or heard any advertising for any brand of alcoholic beverage…on television (Broadcast or Cable); on the radio; in magazines; on billboards or signs; and on the Internet (through webpages, blogs, Facebook, Twitter, Tumblr, etc.)?"

Frequency of Exposure to Alcohol Advertising in Different Media. For each medium in which a participant recalled seeing or hearing alcohol advertising, we asked them to report the frequency with which they saw or heard the advertising: "How often did you see or hear any advertising for any brand of alcohol?" with responses "Less than once a week; Once a week; Several times a week; Once a day; Several times a day." For this analysis, we dichotomized the responses into once a day or more frequently versus less than once a day.

Type of Alcohol Content Seen on the Internet. We identified the type of alcohol content seen on the Internet with the following question: "Which of the following have you seen on the Internet (through webpages, blogs, Facebook, Twitter, Tumblr, etc.)?" Participants were asked the question regarding the following content types: advertisements of alcoholic beverages with pictures or video; pictures or videos of celebrities using alcohol products; pictures or videos of celebrities wearing clothes or other items with an alcoholic beverage logo or name; pictures or videos of celebrities showing the negative effect(s) of using alcohol; articles, news, blogs about alcohol products or merchandise; articles, news, blogs about alcohol brand-sponsored events, parties or concerts; pictures or videos of friends/peers using alcohol products or merchandise; pictures or videos of friends/peers showing the negative effect(s) of using alcohol; pictures or videos of yourself using alcohol products or merchandise; pictures or videos of yourself showing the negative effect (s) of using alcohol.

Engagement with Alcohol Content on the Internet. To determine whether participants had actively engaged with any alcohol content on the Internet, we posed 3 questions: "Which of the following have you liked/shared/posted on the Internet (through webpages, blogs, Facebook, Twitter, Tumblr, etc.)?" The questions were asked for each of the different types of alcohol content outlined above, with response options of "yes," "no," or "not sure."

Analysis

We calculated the unweighted and weighted proportions of the sample based on the following demographic characteristics: age group, sex (male/female), race/ethnicity, annual household income, and region of the United States.

For purposes of comparing youth to adult exposure, we classified all respondents under the legal drinking age (i.e., ages 13 to 20) as "youth" and all others as "adults." The weighted proportion of youth and adults who reported exposure to alcohol advertising was compared by media type and advertising content. We also calculated the proportion of youth and adults who recalled seeing advertisements at least once a day in each medium.

The weighted proportion of adults and youth who reported actively engaging with alcohol advertising on the Internet (i.e., liking, sharing, or posting a specific type of alcohol content) was assessed among those respondents who reported exposure to each type of alcohol content on the Internet.

For all comparisons, chi-square testing was used to assess the statistical significance of differences in proportions between adults and youth ($p \leq 0.05$).

RESULTS

Response Rate and Sample Demographics

The survey response rate varied based on the age of panel members who were invited to participate, and the procedures used to recruit them. A total of 2,376 parents or guardians were contacted to request permission to survey their teen panelists aged 13 to 17 years. Of these, 1,172 parents or guardians (49%) granted permission for their teen to participant in the survey. Of the 1,172 panelists who were granted permission to participate, 773 (66%) chose to participate and completed the survey, resulting in an overall response rate of 32% (773/2,376). Among panelists aged 18 to 20 years, there were 1,003 panel members invited to participate of whom 419 (41%) consented and completed the survey. Among panelists aged 21 or older, a total of 2,113 invitations to participate were sent out and 1,124 (53%) consented and completed the survey. Survey completion time varied considerably as respondents were allowed to start the survey and complete it at their convenience; however, the median survey completion time was 10 minutes for youth respondents and 11 minutes for adults.

The youth (age 13 to 17) respondents had nearly equivalent male and female participation, as did the adults (age 21+). The respondents aged 18 to 20 had a higher proportion of females (n = 255) than males (n = 164). Patterns by race/ethnicity were similar across age groups; nearly 62% of the youth, 59% of the 18- to 20-year-olds and 75% of adults were white, 10, 9, and 8% were black, 18, 23, and 11% were Hispanic, and roughly 10, 10, and 6% reported identifying as another race or as 2+ races, respectively (Table 1). Most youth were from the south (34%), most 18- to 20-year-olds were from the west (33%) followed by the south (30%), and most adults were from the south (36%). The plurality of respondents across age groups fell into the $40,000 to $99,999 annual household income range, and nearly all respondents had access to the Internet.

Self-Reported Youth and Adult Exposure to Alcohol Advertising

Youth recalled greater exposure to alcohol advertising in the past 30 days than adults in every medium with the

Table 1. Characteristics of Survey Participants, GfK Custom Research, September 25 to October 14, 2013

	Youth (13 to 17) n (%)	Young Adults (18 to 20) n (%)	Adults (21+) n (%)
Total	773 (33.4)	419 (18.1)	1,124 (48.5)
Age			
13	188 (8.1)	–	–
14	151 (6.5)	–	–
15	158 (6.8)	–	–
16	122 (5.3)	–	–
17	154 (6.7)	–	–
18 to 20	–	419 (18.1)	–
21 to 34	–	–	246 (10.6)
35 to 49	–	–	273 (11.8)
50+	–	–	605 (26.1)
Gender			
Male	386 (49.9)	164 (39.1)	562 (50)
Female	387 (50.1)	255 (60.9)	562 (50)
Race/Ethnicity			
White, non-Hispanic	477 (61.9)	246 (58.7)	843 (75.0)
Black, non-Hispanic	75 (9.7)	37 (8.8)	86 (7.7)
Hispanic	137 (17.8)	95 (22.7)	125 (11.1)
Other, non-Hispanic	23 (3.0)	20 (4.8)	46 (4.1)
Mixed, non-Hispanic[a]	59 (7.6)	21 (5.0)	24 (2.1)
Region			
Northeast	136 (17.6)	61 (14.6)	205 (18.2)
Midwest	209 (27.0)	94 (22.4)	249 (22.2)
South	263 (34.0)	124 (29.6)	405 (36.0)
West	165 (21.4)	140 (33.4)	265 (23.6)
Annual household income			
<$15,000	66 (8.5)	62 (14.8)	95 (8.5)
$15,000 to $39,999	160 (20.7)	104 (24.8)	241 (21.4)
$40,000 to $99,999	353 (45.7)	156 (37.2)	479 (42.6)
$100,000 or more	194 (25.1)	97 (23.2)	309 (27.5)
Internet access[b]			
Yes	738 (95.5)	399 (95.2)	925 (82.3)
No	35 (4.5)	20 (4.8)	199 (17.7)

[a]Indicates respondents who reported identifying as more than 1 race.
[b]Internet access refers to access prior to joining the Knowledge Networks panel. Recruited panelists who did not have Internet access were provided with access by Knowledge Networks.

Table 2. Weighted Proportion of Reported Exposure to Alcohol Marketing During the Past 30 Days Among Youth and Adults by Media Type, GfK Custom Research, September 25 to October 14, 2013

Media Type	Youth % (95% CI) (n = 1,192)	Adults % (95% CI) (n = 1,124)
Internet	29.7 (26.3, 33.2)	16.8 (14.3, 19.7)***
TV	69.2 (65.7, 72.5)	61.9 (58.3, 65.4)**
Radio	24.8 (21.7, 28.1)	16.7 (14.2, 19.5)***
Magazines	35.7 (32.1, 39.4)	36.4 (33.1, 39.9)
Billboards	54.8 (51.1, 58.3)	35.4 (32.1, 38.8)***

CI, confidence interval.
** $p < 0.01$, *** $p < 0.001$.
Proportions in this table are weighted to be representative of the U.S. population based on census data.

Table 3. Weighted Proportion of Reported Daily Exposure to Alcohol Marketing Among Youth and Adults Exposed During the Past 30 Days by Media Type, GfK Custom Research, September 25 to October 14, 2013

Media Type	Youth % (95% CI)	Adults % (95% CI)
Internet	17.3 (12.3, 23.8)	19.4 (13.2, 27.5)
TV	19.3 (16.0, 23.2)	16.8 (13.7, 20.4)
Radio	14.4 (9.7, 21.1)	11.7 (7.2, 18.3)
Magazines	6.8 (3.9, 11.7)	1.7 (0.7, 4.1)**
Billboards	11.7 (8.6, 15.6)	10.8 (7.7, 15.1)

CI, confidence interval.
** $p < 0.01$.
Proportions in this table are weighted to be representative of the U.S. population based on census data.
Proportions are limited to those who reported exposure to alcohol marketing during the past 30 days by media: Internet youth (n = 329) and adults (n = 178); TV youth (n = 814) and adults (n = 729); radio youth (n = 272) and adults (n = 185); magazines youth (n = 380) and adults (n = 416); billboards youth (n = 637) and adults (n = 411).

exception of magazines (Table 2). Youth age 13 to 20 years were significantly more likely than adults aged ≥21 years to recall seeing alcohol ads on the Internet (29.7% vs. 16.8%, $p < 0.001$), on TV (69.2% vs. 61.9%, $p < 0.01$), on the radio (24.8% vs. 16.7%, $p < 0.001$), and on billboards (54.8% vs. 35.4%, $p < 0.001$).

Reported frequency of exposure to daily alcohol marketing was significantly different between youth and adults only for magazines (Table 3), where more youth reported seeing alcohol advertisements in magazines on a daily basis compared to adults (6.8% vs. 1.7%, $p < 0.01$).

More youth than adults recalled seeing certain alcohol-related online content (Table 4), including alcohol advertisements (40% vs. 25.3%, $p < 0.001$); pictures of celebrities using alcohol (36.1% vs. 20.8%, $p < 0.001$); pictures of celebrities wearing clothes or other items with an alcohol brand's logo or name on it (27.7% vs. 15.9%, $p < 0.001$); pictures of celebrities showing the negative effect(s) of alcohol use (25.4% vs. 21.0%, $p < 0.05$); articles, news, or blogs about alcohol products or merchandise (30.3% vs. 24.6%, $p = 0.02$); and articles, news, or blogs about alcohol brand-sponsored events (32% vs. 27.1%, $p = 0.04$).

Youth also interacted with (liked, shared, or posted) some alcohol-related online content in greater proportions than adults, such as celebrities using alcohol (10.7% vs. 5.7%, $p < 0.01$), celebrities wearing branded items (9.8% vs. 5.4%, $p < 0.01$), pictures of celebrities showing the negative effect (s) of using alcohol (9.3% vs. 4.9%, $p < 0.01$), pictures of their friends or peers using alcohol (14.1% vs. 10.1%, $p = 0.02$), and pictures of friends showing the negative effect (s) of using alcohol (9.4% vs. 4.8%, $p < 0.01$).

DISCUSSION

To our knowledge, this is the first pilot study to assess self-reported youth and adult exposure to alcohol advertising across multiple media venues, including the Internet, as well as active engagement with alcohol advertising and promotional materials on the Internet. Although youth response rates were lower than those of adults, they were still above 30%. The survey proved feasible within the time and funds available, and comparable data appear to have been obtained.

Table 4. Weighted Proportion of Reported Exposure to and Engagement with Internet Alcohol Content Among Youth 13 to 20 Years and Adults ≥21 Years by Type of Exposure, GfK Custom Research, September 25 to October 14, 2013

	Viewed Content		Interacted[a] with Content	
Type of Exposure	Youth % (95% CI) (n = 1,192)	Adults % (95% CI) (n = 1,124)	Youth % (95% CI) (n = 1,192)	Adults % (95% CI) (n = 1,124)
All alcohol advertisements	40.0 (36.4, 43.6)	25.3 (22.3, 28.5)***	9.7 (7.6, 12.5)	7.1 (5.3, 9.4)
Celebrities using alcohol	36.1 (32.6, 39.7)	20.8 (18.0, 23.8)***	10.7 (8.5, 13.5)	5.7 (4.0, 8.0)**
Celebrities wearing alcohol-branded items	27.7 (24.5, 31.3)	15.9 (13.5, 18.7)***	9.8 (7.7, 12.4)	5.4 (3.7, 7.7)**
Celebrities showing negative effects of alcohol use	25.4 (22.3, 28.8)	21.0 (18.2, 24.1)*	9.3 (7.1, 12.0)	4.9 (3.4, 7.0)**
Articles, news or blogs about alcohol	30.3 (26.9, 33.9)	24.6 (21.6, 27.7)*	8.2 (6.2, 10.8)	6.7 (5.1, 8.9)
Articles, news or blogs about sponsored events	32.0 (28.6, 35.6)	27.1 (24.1, 30.3)*	9.4 (7.3, 12.1)	6.7 (5.1, 8.8)
Friends/peers using alcohol	29.5 (26.2, 32.9)	30.6 (27.5, 33.9)	14.1 (11.6, 17.0)	10.1 (8.1, 12.4)*
Friends/peers showing negative effects of alcohol use	16.1 (13.5, 19.0)	13.6 (11.3, 16.2)	9.4 (7.4, 12.0)	4.8 (3.4, 6.8)**
Pictures of yourself using alcohol	8.1 (6.1, 10.6)	10.6 (8.6, 12.9)	6.3 (4.5, 8.8)	8.9 (7.0, 11.2)
Pictures of yourself showing negative effects of alcohol use	5.3 (3.7, 7.5)	4.3 (3.0, 6.3)	4.4 (3.0, 6.5)	3.1 (2.0, 4.8)

CI, confidence interval.
[a]Interacting with content included liking, sharing, or posting the content online.
$^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.
Proportions in this table are weighted to be representative of the U.S. population based on census data. Respondents reporting "not sure" were counted as not having been exposed to content.

Overall, youth were significantly more likely than adults to report past 30-day exposure to alcohol marketing on most media, including on the Internet, radio, billboards, and television, but less likely to report this exposure in magazines. These differences were most apparent on the Internet, where youth were almost twice as likely as adults to report being exposed to alcohol advertising during the past 30 days. Youth were also more likely than adults to interact with that content, including liking or sharing alcohol content with others, particularly content involving alcohol use or the wearing of alcohol-branded items by celebrities.

Although the purpose of this paper was to compare youth and adult exposure, it is interesting that within the youth population itself, monthly exposure to traditional media—radio, television and billboards—was more prevalent than exposure on the Internet; however, regarding daily exposure, differences in prevalence between these media and the Internet were nonsignificant. These findings may have important implications for future monitoring of overall youth exposure to alcohol marketing.

The greater self-reported exposure of youth to alcohol marketing on the Internet is consistent with the greater use of online media, and particularly emerging social media such as Twitter, by youth than adults (Madden et al., 2013). Conversely, the lower past 30-day exposure to alcohol advertising in magazines is consistent with the findings of studies showing that young people are less likely to spend time reading magazines (Rideout et al., 2010). However, the findings of this study contrast with those of prior studies of youth and adult exposure to alcohol advertising based on commercially available data reporting the placement of alcohol advertisements, which have found that youth are *not* more exposed to alcohol advertising on television and radio programs than adults overall (Center on Alcohol Marketing and Youth, 2010, 2011). Recent research has found that from July 2013 through June 2015, advertising on cable TV programming with youth audiences that exceeded the alcohol industry's voluntary placement standards generated 3.8 billion youth alcohol advertising impressions. Most of these youth impressions occurred on cable TV programs that had previously resulted in noncompliant exposure, emphasizing the need to further reduce the placement of alcohol advertising in media that overexpose youth (Ross et al., 2016).

The higher self-reported exposure to alcohol advertising on TV among youth relative to adults in this study may be due, at least in part, to the fact that alcohol advertising is more salient—that is, generates greater appreciation and awareness—for youth compared to adults (Aitken et al., 1988; Kelly and Edwards, 1998). In fact, there is a large body of research showing that youth may be particularly vulnerable to images and messages that are commonly used in alcohol advertising (see, e.g., Austin and Hust, 2005; Chen et al., 2005; Pechmann et al., 2005). Youth are also going through a process of identity formation and are more likely than children and adults to identify with and model the actions of commercial actors (Martin and Kennedy, 1993). Because youth are still developing executive control, they may also be more susceptible to alcohol advertisements with certain stylistic features such as animation, frequent camera cuts, and loud and fast music. These features have been shown to enhance adult recall through triggering an involuntary, high level of attention paid to the stimuli (Lang, 2000; Lang et al., 2000; Morgan et al., 2003; Nierdeppe et al., 2007; Pechmann et al., 2005). In addition, youth may have low advertising literacy, thus making it difficult for them to critically evaluate persuasive messaging in alcohol advertisements (Boush, 2001; Rozendaal et al., 2011).

Furthermore, alcohol advertising increasingly relies on lifestyle-oriented appeals over product-oriented appeals (Casswell, 1995), and youth exposure to alcohol advertising has been shown to correlate with youth expectancies that alcohol use is part of a fulfilling life, that it leads to happiness, fun, and social acceptance (Fleming et al., 2004; Jones and Donovan, 2001; Sargent et al., 2006; Wills et al., 2009). Taken together, these vulnerabilities and expectancies may increase the salience of alcohol ads for youth under age 21 years and thus make it more likely that they will recall this advertising than adults even if the amount of advertising they are exposed to is not significantly different.

In 2014, the Federal Trade Commission (FTC) published the most recent of a series of reports on advertising self-regulation by the U.S. alcohol industry and included an assessment of the effectiveness of self-regulation in shielding youth from undue exposure to alcohol marketing (Federal Trade Commission, 2014). The FTC assessed the industry's compliance with the voluntary standard of not placing advertising and marketing materials before audiences with <71.6% adult audiences, and noted that 76% of Facebook users, 78% of registered YouTube users, and 79% of Twitter users are age 21 or older. Based on reported audience composition, the FTC concluded that alcohol advertising on these sites was in compliance with self-regulatory guidelines. However, estimates of audience composition cannot account for the individual viewing experiences of youth exposed to alcohol advertising through these media channels, which could significantly increase the salience of alcohol advertisements for exposed youth. That youth reported higher levels of exposure to alcohol advertising than adults in every medium except magazines suggests that current alcohol industry standards may not be sufficiently protective against youth exposure.

This study has several limitations. First, survey response rates were low, particularly for youth, falling just below Knowledge Network's expected range (33 to 41%) for youth responses to their panel surveys (Wendy Mansfield, GfK, e-mail to David Jernigan, July 30, 2015), and no information was available on the characteristics of nonrespondents. It is therefore unknown whether survey participants were representative of youth in the general population in the United States. Second, although weights were used to adjust for various demographic characteristics as well as nonresponse to the survey, these weights did not account for differences in alcohol consumption that could have affected a respondent's recall of alcohol advertising. Third, as previously noted, youth and adult recall of exposure to alcohol marketing may differ based on the differential salience of or experience with the alcohol marketing to which they were exposed, which might have increased self-reported exposure to alcohol advertising by youth relative to adults. Fourth, the survey questions used in this study have not previously been used to assess advertising exposure and may not have been fully understood by some survey respondents. However, the questions were pilot-tested by Knowledge Networks among a subgroup of respondents prior to survey administration, and no major difficulties were reported in respondents' ability to understand and respond to the questions.

There are several policies used by providers of Internet services to reduce youth exposure to alcohol marketing in digital and social media. These include restricting youth access to alcohol company-sponsored pages on Facebook by only allowing Facebook users who are age 21 or older to access them (a practice known as "age-gating"); requiring a user to affirm that they are age 21 years or older before allowing them to follow an alcohol brand feed on Twitter; requiring a potential viewer of the "channels" of branded content that alcohol companies maintain on YouTube to affirm that they are of legal drinking age; and voluntary guidelines used by alcohol companies to limit the placement of alcohol content to webpages where adults are likely to comprise 70% or more of the viewers (Global Alcohol Producers Group, 2014). However, these policies are difficult to enforce, and it is therefore unclear what impact they have on reducing youth exposure to alcohol advertising on the Internet (Jernigan and Rushman, 2014). In fact, about 40% of teenagers responding to a survey on Internet use have reported lying about their age to access a website or sign up for an online account (Madden et al., 2013).

CONCLUSION

Youth report higher levels of exposure to alcohol marketing placed in most media than adults, particularly alcohol marketing placed on the Internet. This is concerning given the known relationship between youth exposure to alcohol marketing and the initiation of alcohol consumption as well more hazardous alcohol consumption by those who drink. Furthermore, youth report being more likely to actively respond to Internet-based advertising of alcoholic beverages than adults, suggesting a higher level of engagement with this advertising, which appears to have been associated with an increase in brand-specific alcohol sales (Diageo plc, 2011).

As alcohol companies increase their spending on alcohol marketing on the Internet and in social media (Federal Trade Commission, 2014), the public health field is challenged to develop and deploy new methods for monitoring youth and adult exposure to marketing in these media. Nielsen, Inc, has also called for greater cooperation across media platforms to facilitate more accurate measurement of total marketing exposure (Connor, 2015) and is itself offering new data systems that "watermark" Internet advertising content to facilitate tracing its impact.

The findings of this study emphasize the need to improve the monitoring of youth exposure to alcohol advertising in both traditional and digital media, and in particular to conduct independent monitoring of compliance with voluntary industry standards on the placement of alcohol advertising, to reduce noncompliant alcohol advertising exposure, and to continue to assess the effectiveness of current voluntary standards in protecting against youth overexposure. Additional research is also needed to assess the impact of youth exposure to and engagement with alcohol marketing in digital and social media on underage drinking and related harms.

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